S6 Proofs for the threshold certificate

This section proves the lemmas that Appendix G summarizes, and so completes the proof of Theorem 4.4. We use the notation of that appendix, and in particular the inequality (9). We write $\Phi_N(t) = \sum_{n\le N} c_n (it)^n$ for the partial sums of the moment series.

S6.A One value of the memory

Lemma S6.1 (moment bounds)**.** *Let $N \ge 3$.*

(a) Put $A = ((N+1)a - 1)/N$. If $m_j \le AB^j$ for $1 \le j \le N$, then $m_n \le AB^n$ for all $n \ge 1$.

(b) Let $S \subset \mathbb{C}$ be compact, with $\frac12 < \alpha \le \operatorname{Re} z$, $|\operatorname{Im} z| \le \rho$ and $|z| \le 1$ for $z \in S$. Define $m_n(z)$ by the recursion with z in place of a, and $c_n(z) = m_n(z)/\Gamma(1+nz)$. Then every m_n is analytic near S. Put $A = ((N+1)\alpha - 1)/N$. If $|m_j| \le AB^j$ on S for $1 \le j \le N$, then for all $n \ge 1$ and $z \in S$

$$|m_n(z)| \le AB^n, \qquad |c_n(z)| \le \frac{\tilde A \tilde B^n}{\Gamma(1+n\alpha)}, \qquad \tilde A = Ae^{\rho^2/(2\alpha^2)}, \quad \tilde B = Be^{\rho^2/(2\alpha)}.$$

Proof. For (a) we use induction on $n > N$. As $\theta_j \le 1$ and $na - \theta_n \ge na - 1 > 0$, the bound for smaller indices gives $m_n \le (n-1)A^2B^n/(na-1)$. The ratio $(na-1)/(n-1)$ increases in n, since its derivative is $(1-a)/(n-1)^2$, and it equals A at $n = N+1$. So $m_n \le AB^n$.

For (b), $\operatorname{Re}(nz - \theta_n) \ge n\alpha - 1 > 0$ on S, because $\alpha \le 1$. So m_n is analytic near S, and the induction of (a) goes through with $|\theta_j| \le 1$. For $x > 0$ and real y,

$$\Big|\frac{\Gamma(x)}{\Gamma(x+iy)}\Big|^2 = \prod_{k\ge0}\Big(1 + \frac{y^2}{(x+k)^2}\Big) \le \exp\Big(\frac{y^2}{x^2} + \frac{y^2}{x}\Big)$$

by [42, 5.8.3]. We take $x = 1 + n\operatorname{Re} z \ge 1 + n\alpha$ and $|y| \le n\rho$. Then $y^2/x^2 \le \rho^2/\alpha^2$ and $y^2/x \le n\rho^2/\alpha$. Finally $\Gamma(1 + n\operatorname{Re} z) \ge \Gamma(1+n\alpha)$, as Γ increases on $[1.47, \infty)$. □

Lemma S6.2 (truncation)**.** *Let $0 < \alpha \le 1$, and suppose that $|c_n| \le \tilde A \tilde B^n/\Gamma(1+n\alpha)$ for $n > N$. Let $t > 0$, $0 \le k \le N$ and $n_0 = N+1$, and put*

$$u = \frac{\tilde A \tilde B^{n_0} n_0^k t^{n_0-k}}{\Gamma(1+\alpha n_0)}, \qquad r = \tilde B t\Big(1 + \frac{1}{n_0}\Big)^k \frac{\Gamma(1+\alpha n_0)}{\Gamma(1+\alpha n_0+\alpha)}.$$

If $r < 1$, then $\big|\frac{d^k}{dt^k}\sum_{n>N} c_n(it)^n\big| \le \tau_k(t) := u/(1-r)$, and τ_k increases on $(0,t]$.

Proof. The kth derivative of $(it)^n$ has modulus at most $n^k t^{n-k}$. The ratio of consecutive terms of the majorant $\tilde A \tilde B^n n^k t^{n-k}/\Gamma(1+\alpha n)$ is $\tilde B t(1+1/n)^k \Gamma(1+\alpha n)/\Gamma(1+\alpha n+\alpha)$. It decreases in n, because $\log\Gamma$ is convex. So the majorant is at most a geometric series with first term u and ratio r. Both u and r increase in t. □

Lemma S6.3 (cells)**.** *Let ν be even, $t_c > 0$ and $|\delta| \le h$. Then $|\phi(t_c+\delta)| \le \sum_{k<\nu} |\phi^{(k)}(t_c)| h^k/k! + c_\nu h^\nu$.*

Proof. This is Taylor's formula, with $|\phi^{(\nu)}| \le \mathbb{E}|L|^\nu = \mathbb{E}L^\nu = \nu!\, c_\nu$. □

Lemma S6.4 (the tail)**.** *Let $0 < T_2 < T_3$ with $T_3 \ge 2^a T_2$, and let $\beta > 3$ with $a\beta > 1$. Cover $[0, T_3]$ by cells $[l_j, r_j]$ with $l_0 = 0$ and with T_2 among the end points, and let $s_j \ge \sup_{[l_j, r_j]} |\phi|$. Let C be the largest $s_j r_j^\beta$ over the cells that meet $[T_2, T_3]$, and put*

$$R = T_2^{-1/a} \sum_{r_j \le T_2} s_j \Big(1 - \frac{1}{2}\Big(\frac{r_j}{T_2}\Big)^{1/a}\Big)^{-a\beta} \big(r_j^{1/a} - l_j^{1/a}\big) + \frac{2^{a\beta} C T_2^{-\beta}}{a\beta - 1}.$$

If $R < 1$, then $|\phi(t)| \le C t^{-\beta}$ for every $t \ge T_3$.

Proof. Let $b = s_j$ on the cells in $[0, T_3)$ and $b(t) = Ct^{-\beta}$ for $t \ge T_3$. On a cell that meets $[T_2, T_3]$ we have $s_j \le C r_j^{-\beta} \le C t^{-\beta}$. So $b(t) \le Ct^{-\beta}$ for $t \ge T_2$, and $|\phi| \le b$ on $[0, T_3]$. Let $J(t) = \int_0^1 b(tv^a)\, b(t(1-v)^a)\, dv$.

We show that $J(t) \le R b(t)$ for $t \ge T_3$. By symmetry $J(t)$ is twice the integral over $v \le \frac12$. There $t(1-v)^a \ge 2^{-a} T_3 \ge T_2$, so $b(t(1-v)^a) \le b(t)(1-v)^{-a\beta}$. Hence $J(t)/b(t)$ is at most $2\int_0^{1/2} b(tv^a)(1-v)^{-a\beta}\, dv$. Put $v_0 = (T_2/t)^{1/a} \le \frac12$. On $v < v_0$ we substitute $u = tv^a$, and this part becomes

$$\frac{2}{a} t^{-1/a} \int_0^{T_2} s(u)\big(1 - (u/t)^{1/a}\big)^{-a\beta} u^{1/a - 1}\, du.$$

Here $t^{-1/a} \le \frac12 T_2^{-1/a}$ and $(u/t)^{1/a} \le \frac12 (u/T_2)^{1/a}$. The weight increases in u, so on each cell we take it at r_j. As $\int_l^r u^{1/a-1} du = a(r^{1/a} - l^{1/a})$, this part is at most the first term of R. On $v \ge v_0$ we use $b(tv^a) \le Ct^{-\beta} v^{-a\beta}$ and $(1-v)^{-a\beta} \le 2^{a\beta}$. This part is at most $2^{1+a\beta} T_2^{-\beta} (T_2/t)^{1/a} C/(a\beta - 1)$, which is at most the second term of R.

Let ς be the supremum of the $\tau \ge T_3$ with $|\phi| \le b$ on $[0, \tau]$, and suppose that $\varsigma < \infty$. By continuity $|\phi| \le b$ on $[0, \varsigma]$. The arguments in (9) at $t = \varsigma$ lie in $[0, \varsigma]$, so $|\phi(\varsigma)| \le J(\varsigma) \le R b(\varsigma) < b(\varsigma)$. As $|\phi|$ and b are continuous on $[T_3, \infty)$, we get $|\phi| < b$ near ς, against the choice of ς. □

This is an explicit form of Step 4 in the proof of [8, Theorem 2.5], with computed constants.

Lemma S6.5 (the second derivative)**.** *If $|\phi(t)| \le Ct^{-\beta}$ for $t \ge T_3$, with $\beta > 3$, then g_a has two continuous derivatives and*

$$g_a''(0) = -\frac{1}{\pi}\Big(\sum_{\substack{n \le N \\ n \text{ even}}} (-1)^{n/2} c_n \frac{T_3^{n+3}}{n+3} + E\Big), \qquad |E| \le \frac{T_3^3}{3}\tau_0(T_3) + \frac{C T_3^{3-\beta}}{\beta - 3}.$$

Proof. As $t^2 \operatorname{Re}\phi(t)$ is integrable, Fourier inversion gives $g_a(x) = \frac{1}{2\pi} \int e^{-itx} \operatorname{Re}\phi(t)\, dt$, and dominated convergence allows two derivatives under the integral [40, Section 3.3]. So $g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2 \operatorname{Re}\phi(t)\, dt$. On $[0, T_3]$ we replace ϕ by Φ_N, whose real part is $\sum_{n \text{ even}} (-1)^{n/2} c_n t^n$, and integrate term by term. By Lemma S6.2 the error is at most $\int_0^{T_3} t^2 \tau_0(t)\, dt \le \frac13 T_3^3 \tau_0(T_3)$. The integral over $t > T_3$ is at most $C T_3^{3-\beta}/(\beta - 3)$. □

S6.B Intervals of the memory

Ball arithmetic with a as an interval fails here. Near the threshold the series for ϕ cancels terms of size about 10^{60}, so the boxes in a would have to be narrower than about 10^{-65}. We expand in a around one exact point instead.

Lemma S6.6 (Cauchy remainder). *Let h be analytic near the closed disc $|z-a_0|\le\rho$, with $|h|\le M$ there and Taylor coefficients h_i at a_0. If $0<q<1$ and $|a-a_0|\le q\rho$, then $|h(a)-\sum_{i\le K}h_i(a-a_0)^i|\le Mq^{K+1}/(1-q)$.*

Proof. Cauchy's estimate gives $|h_i|\le M\rho^{-i}$. Sum the geometric tail. □

Let $I=[a_0-w,a_0+w]$, $0<q<1$ and $\rho=w/q$, and let S be the square $a_0+[-\rho,\rho]+i[-\rho,\rho]$, which contains the disc of Lemma S6.6. Suppose that $\alpha=a_0-\rho>\frac12$, that $|z|\le1$ on S, and that $\tilde A,\tilde B$ are as in Lemma S6.1(b). Let τ_k be the bound of Lemma S6.2 with these constants. It holds for every real $a\in I$. For a cell with centre t_c write $\partial_t^k\Phi_N(z,t_c)=E_k(z)+iO_k(z)$, where E_k is the sum over even n and iO_k the sum over odd n. Both E_k and O_k are real for real z. Let $E_{k,i}$ and $O_{k,i}$ be their Taylor coefficients at a_0, and put $M_k=\sum_{n\le N}\sup_S|c_n|\,n(n-1)\cdots(n-k+1)\,t_c^{n-k}$.

Proposition S6.7 (enclosure on an interval). *(a) For every $a\in I$ and every t in the cell with centre t_c and half-width h,*

$$|\phi_a(t)|\le\sum_{k<12}\Big(\sum_{i\le K}w^i\sqrt{E_{k,i}^2+O_{k,i}^2}+\frac{M_kq^{K+1}}{1-q}+\tau_k(t_c)\Big)\frac{h^k}{k!}+h^{12}\sup_S|c_{12}|.$$

(b) Let T_2, $T_3\ge2^{a_0+w}T_2$, β and the cells be as in Lemma S6.4, with s_j the bound of (a), and suppose that $R<1$ for every $a\in I$. Let $F(z)=\sum_{n\le N,\,n\ even}(-1)^{n/2}c_n(z)T_3^{n+3}/(n+3)$, with Taylor coefficients F_i at a_0, and let M_F be the same sum with $\sup_S|c_n|$ in place of $(-1)^{n/2}c_n(z)$. Then for every $a\in I$

$$\Big|g_a''(0)+\frac1\pi\sum_{i\le K}F_i(a-a_0)^i\Big|\le\eta=\frac1\pi\Big(\frac{M_Fq^{K+1}}{1-q}+\frac{T_3^3}{3}\tau_0(T_3)+\frac{CT_3^{3-\beta}}{\beta-3}\Big),$$

and $a\mapsto g_a''(0)$ is continuous on I.

Proof. For (a) we apply Lemma S6.3 with $\nu=12$, and Lemma S6.2 to replace $\phi_a^{(k)}$ by $\partial_t^k\Phi_N$. For real a we have $|\partial_t^k\Phi_N(a,t_c)|=|(E_k(a),O_k(a))|$. Lemma S6.6 for E_k and O_k, with $\sup|E_k|+\sup|O_k|\le M_k$ on S, and the triangle inequality in $\mathbb{R}^2$ give the bracket. Also $c_{12}(a)\le\sup_S|c_{12}|$.

For (b), $T_3\ge2^aT_2$ for every $a\in I$, so Lemma S6.4 applies at each a. Then Lemma S6.5 and Lemma S6.6 for F give the enclosure. By Lemma S6.1(b) the series for $\phi_a(t)$ converges uniformly for $a\in I$ and t in compact sets. So $\phi_a(t)$ is continuous in a. Moreover $t^2|\phi_a(t)|$ is at most t^2 on $[0,T_3]$ and $Ct^{2-\beta}$ beyond, uniformly in $a\in I$. Dominated convergence gives the continuity. □

Near $a=\frac12$ the bounds of Lemma S6.1 degenerate, since $m_2=a/(2a-1)$. We rescale.

Lemma S6.8 (the scaled variable). *Put $y=2a-1$ and $\mu_n=m_ny^{\lfloor n/2\rfloor}$.*

(a) We have $\mu_0=\mu_1=1$, $\mu_2=a$, and for $n\ge3$

$$(na-\theta_n)\mu_n=\sum_{j=1}^{n-1}\theta_jy^{e_{nj}}\mu_j\mu_{n-j},$$

where $e_{nj}=1$ if n is even and j is odd, and $e_{nj}=0$ otherwise. Here $na-\theta_n$ is $(n-1)a$ for odd n and $na-1$ for even n. So every μ_n is a rational function of a with no pole in $\operatorname{Re}a>\frac14$.

(b) Let $\sigma^2 = \mathbb{E}L^2$, $X = L/\sigma$ *and* $\hat{c}_n = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor}/\Gamma(1+an)$. *Then* $\sigma^2 = 1/(y\Gamma(2a))$,

$$\mathbb{E}e^{iuX} = \sum_{n \text{ even}} \hat{c}_n (iu)^n + \sqrt{y\Gamma(2a)} \sum_{n \text{ odd}} \hat{c}_n (iu)^n, \qquad g_a''(0) = -\frac{1}{\pi\sigma^3} \int_0^\infty u^2 \operatorname{Re} \mathbb{E}e^{iuX}\, du.$$

Proof. Multiply the recursion for m_n by $y^{\lfloor n/2 \rfloor}$. The exponent $\lfloor n/2 \rfloor - \lfloor j/2 \rfloor - \lfloor (n-j)/2 \rfloor$ is 1 when n is even and j is odd, and 0 otherwise. Also $m_2 = a/y$. This proves (a), as the zeros of the factors are $a = 0$ and $a = 1/n \le \frac{1}{4}$. For (b), $\sigma^2 = 2m_2/\Gamma(1+2a) = 1/(y\Gamma(2a))$ and $\mathbb{E}X^n/n! = c_n \sigma^{-n} = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor} (y\Gamma(2a))^{(n \bmod 2)/2}/\Gamma(1+an)$. The formula for $g_a''(0)$ follows from the substitution $t = u/\sigma$ in Lemma S6.5. □

The fixed-point equation is linear, so X satisfies it too, and (9) and Lemmas S6.3 to S6.5 hold for X. The proof of Lemma S6.1(b) carries over to μ_n for $n \ge 3$ and $\alpha > \frac{1}{4}$, with two changes. The constant A becomes $((N+1)\alpha - 1)/(N\kappa)$, where κ is the largest value of $\max(1, |z|) \max(1, |2z-1|)$ on S. And $\tilde{B}$ gains the factor $\max(1, \sup_S |\Gamma(2z)|)^{1/2}$. For real a the odd part carries the factor $\sqrt{y\Gamma(2a)}$. In the cell bounds we treat the even and odd parts as separate analytic functions. We multiply the odd part by $f = \sup_I \sqrt{y\Gamma(2a)}$ only for real a, and the remainders by $\max(1, f)$. The real part of $\mathbb{E}e^{iuX}$ involves only the even part. So the disc may cross $a = \frac{1}{2}$, and the cover reaches $a = \frac{1}{2}$. The sign of $g_a''(0)$ is the opposite of the sign of the integral, since $\sigma^3 > 0$ and the Fourier formula has a minus sign.

S6.C From the runs to the theorem

The script accepts an interval of a only when $G \pm \eta$ has one sign on each piece of a subdivision of it. In the 7 scaled runs every piece of every interval is negative, so $g_a''(0) < 0$ on $(\frac{1}{2}, 0.668]$. The zero run has 8 intervals that cover $[0.668, 0.6712]$. On the 2 intervals that cover $[0.668, 0.6688]$ every piece is negative. On the 5 intervals that cover $[0.6692, 0.6712]$ every piece is positive. On $[0.6688, 0.6692]$ the pieces up to z_- are negative, the pieces from z_+ on are positive, and only (z_-, z_+) is undecided. Proposition S6.7(b) turns these signs into $g_a''(0) < 0$ on $(\frac{1}{2}, z_-]$ and $g_a''(0) > 0$ on $[z_+, 0.6712]$. The order matters. A positive interval left of z_- would give another zero, and parts (a) and (c) of Theorem 4.4 would fail. So the script prints the sign of each accepted interval with its exact end points. Its mode `zero` fails unless the signs are $-$ before the undecided piece and $+$ after it. Its mode `check` requires the same of the stored records, together with the sign $-$ on every interval of the scaled runs.

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